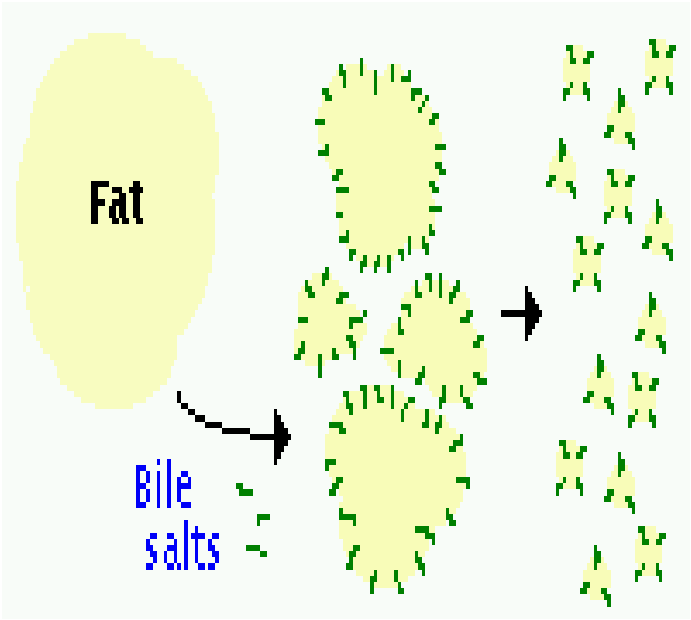
Beta oxidation

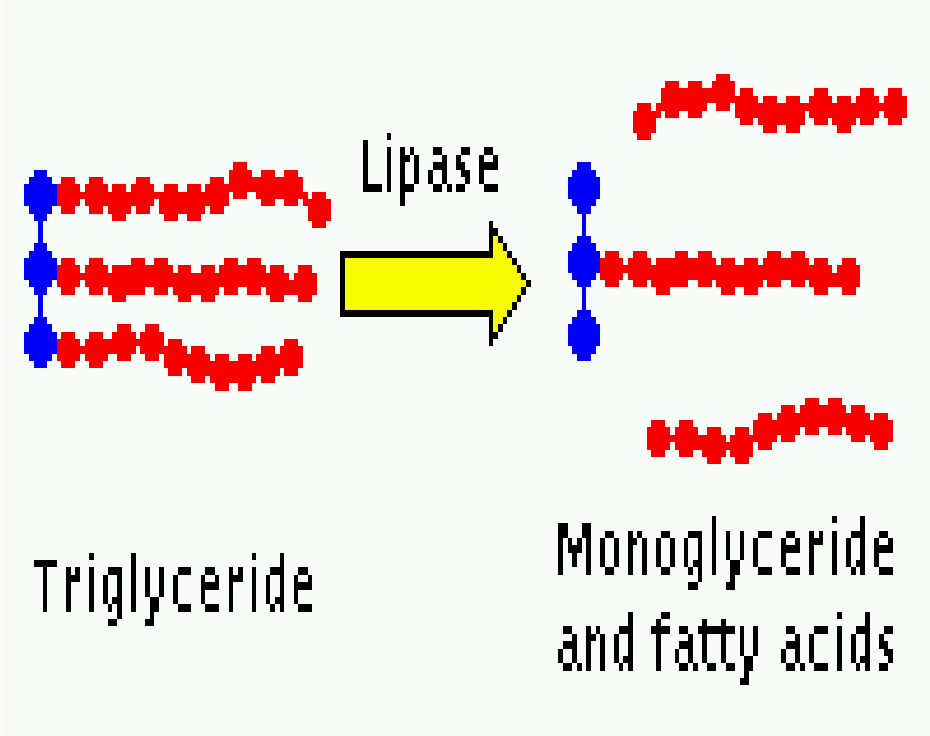
Digestion of dietary lipids

- Occurs in GIT with the action of lipase
- Gastric lipase requires neutral pH for digestion
- In adults much of the lipids pass on unchanged to SI
- Pancreatic lipase converts TGs to FFA + MAG
- Pancreatic cholesteryl esterase → CE to C & FFA
- PL are hydrolysed by phospholipase A2
- Bile salts lower surface tension. Emulsify lipids
- Facilitate the action of enzymes

- Bile salts = synthesised from cholesterol in the liver
- Stored in gall bladder
- Released into intestine
- Lower surface tension
- Help emulsify the fat



- Pancreatic lipase



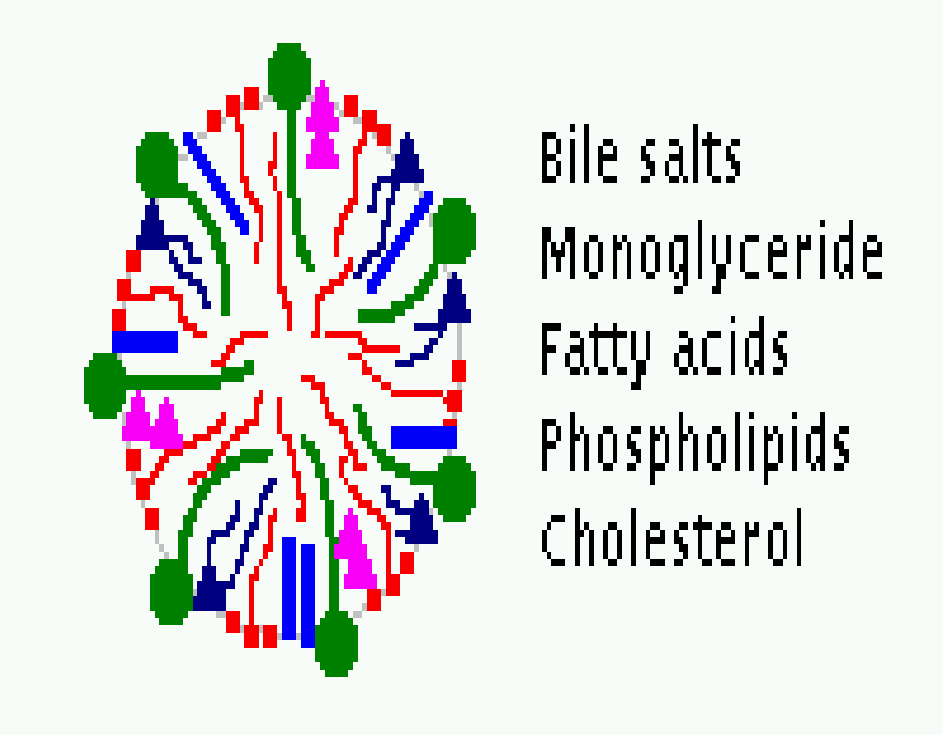
Absorption of lipids across intestinal mucosa

- At the end of lipid digestion

  FFA, MAG, C, BS and lysophospholipids (present in the intestinal lumen)→ micelles

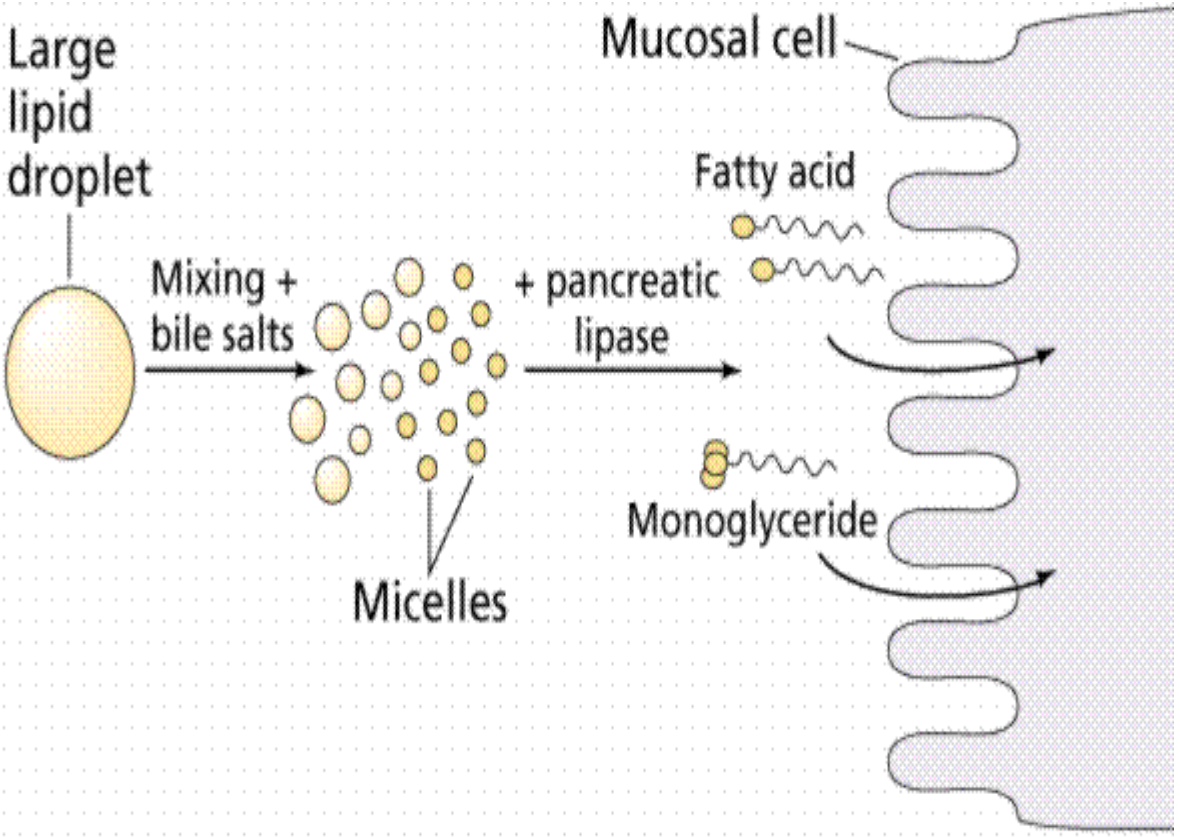
- Micelles pass into the mucosal cells

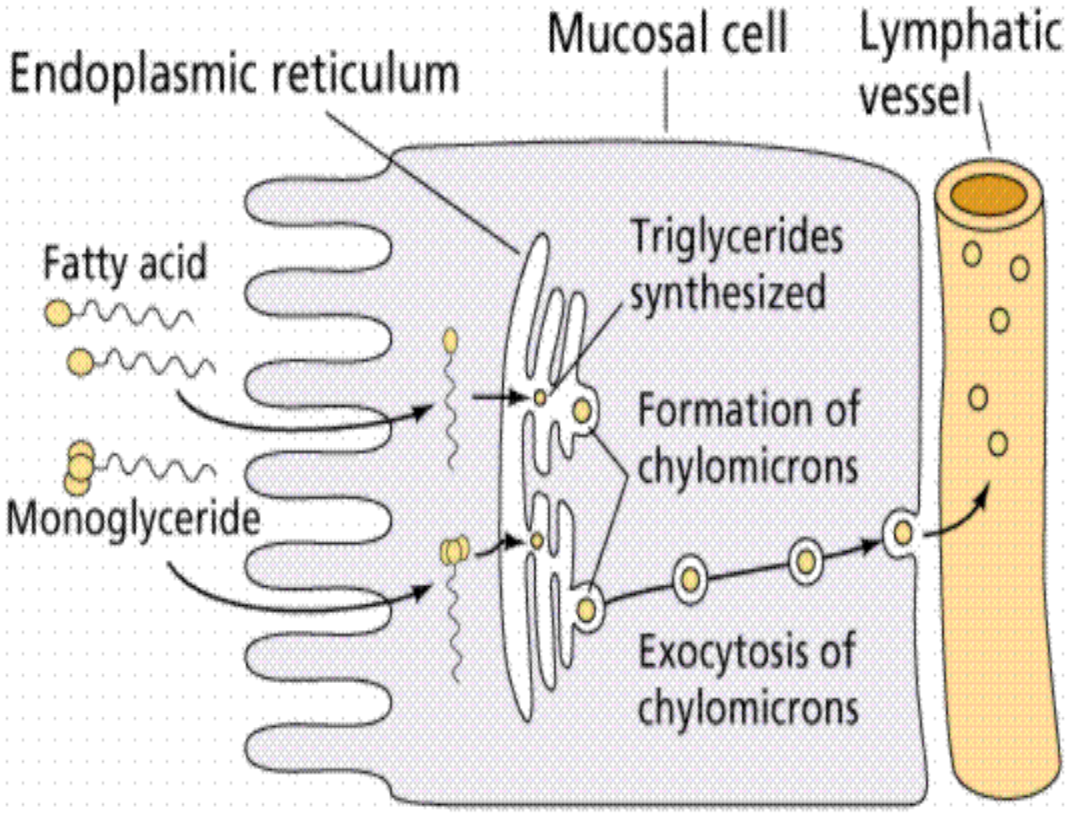
- Micelles
- Amphipathic structures with hydrophobic core containing water insoluble lipids and hydrophilic outer shell soluble in aqueous medium
- Micelles are absorbed across intestinal mucosal cells



Lipid transport from intestinal mucosa

- Once inside the int. mucosal cells, the absorbed lipids are resynthesised to form TGs and CE
- Intestinal cells also synthesise apolipoprotein B48, and package TGs and CE to form chylomicrons
- Chylomicrons are secreted first into the lymphatics and then into the blood


Large
lipid
droplet
Mixing +
bile salts
Micelles
+ pancreatic
lipase
Mucosal cell
Fatty acid
Monoglyceride


Endoplasmic reticulum
Mucosal cell
Lymphatic vessel
Fatty acid
Monoglyceride
Triglycerides synthesized
Formation of chylomicrons
Exocytosis of chylomicrons

- Chylomicrons

- **Chylomicrons** are large lipoprotein particles

- consist of triglycerides (85-92%), phospholipids (6-12%), cholesterol (1-3%) and proteins (1-2%).

- They transport dietary lipids from the intestines to other locations in the body.

- They are one of the five major groups of lipoproteins (chylomicrons, VLDL, IDL, LDL, HDL) that enable fats and cholesterol to move within the water-based solution of the bloodstream.

Chylomicron

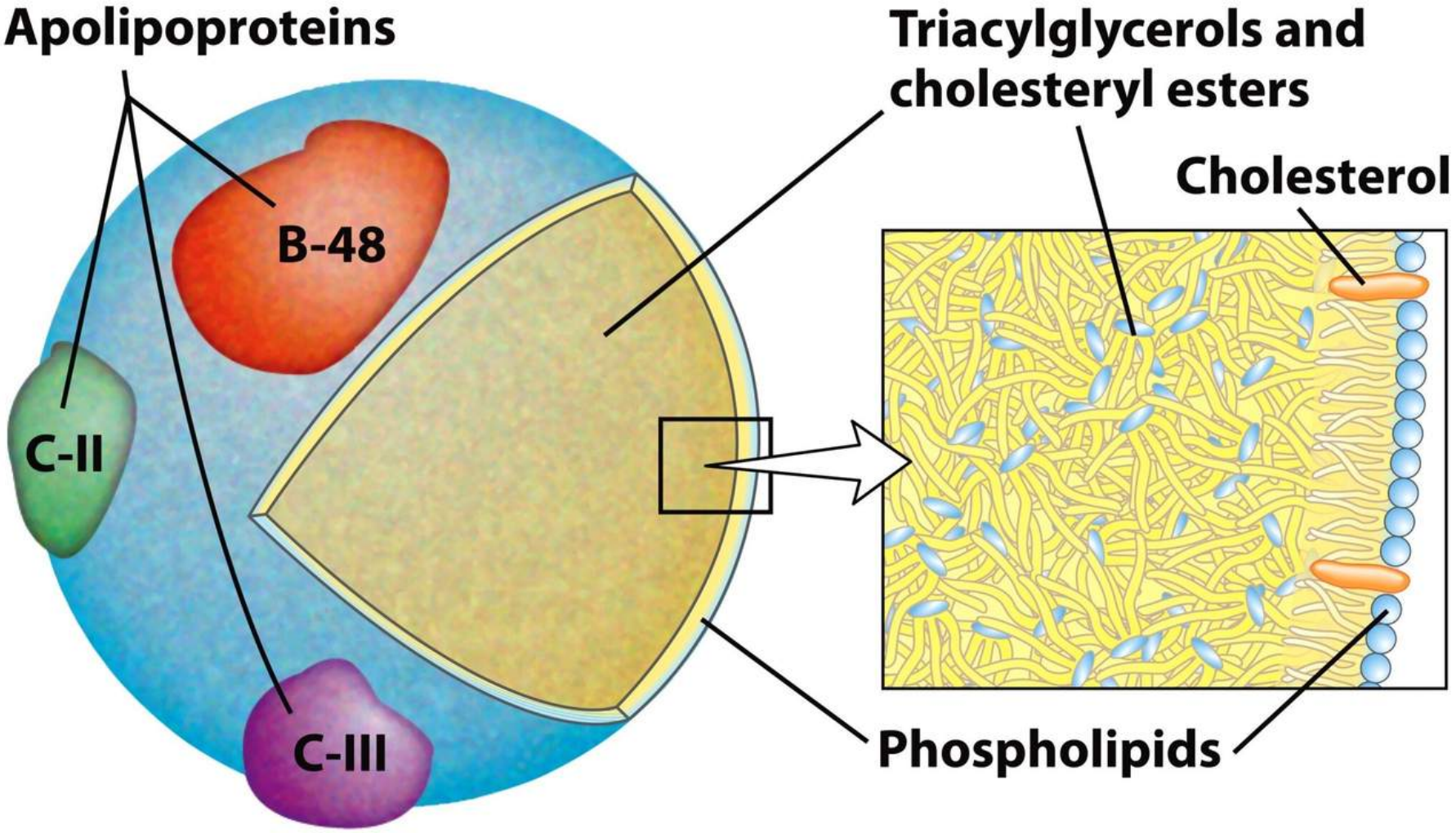


Figure 17-2
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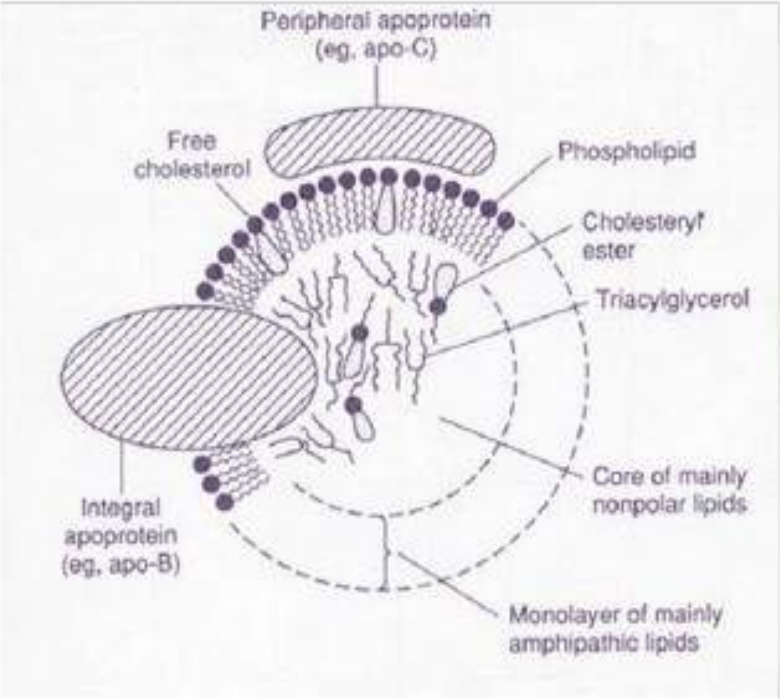
Lipoproteins

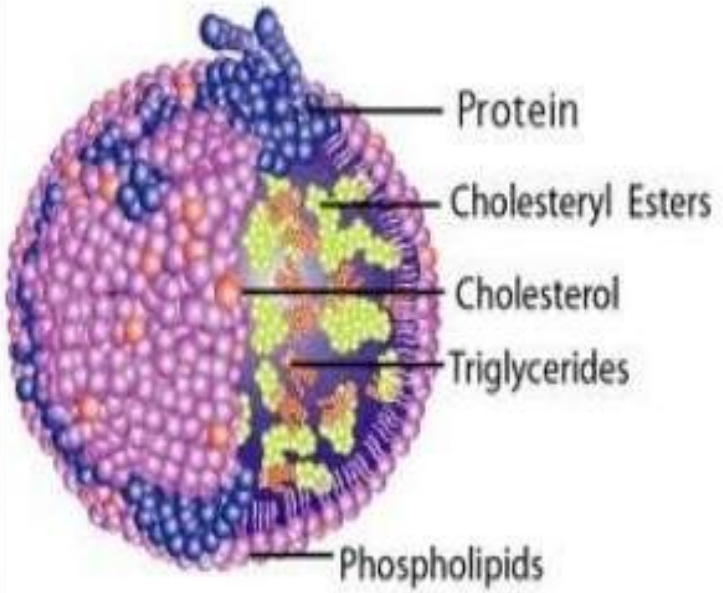
- TGs and C are transported in blood as lipoproteins
- According to densities the LPs are classified as:
- Chylomicrons
- VLDL
- IDL
- LDL
- HDL

Classes of lipoproteins with associated apolipoproteins

Lipoprotien	Function	Apolipoprotein
Chylomicron	Transport dietary TG and CE from intestine to tissues	Apo B48 Apo CII and apo E
VLDL	Transport TG from liver to tissues	B100, CI,CII, CIII
IDL	Picks up cholesterol from HDL to become lDL picked up by liver	B100, E
LDL	Delivers cholesterol from liver	B100
HDL	Picks up cholesterol accumulating in blood vessels	AI, AII, AIV, CI, CII,CIII, D, E

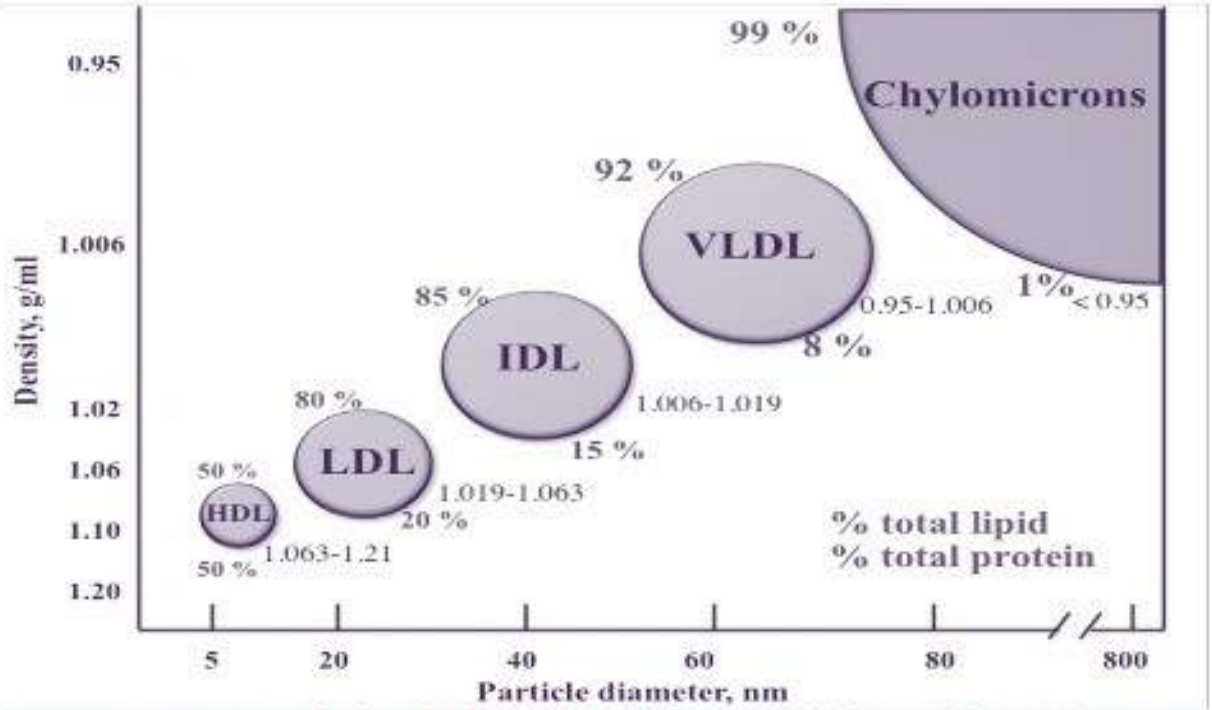
General Structure of Lipo proteins





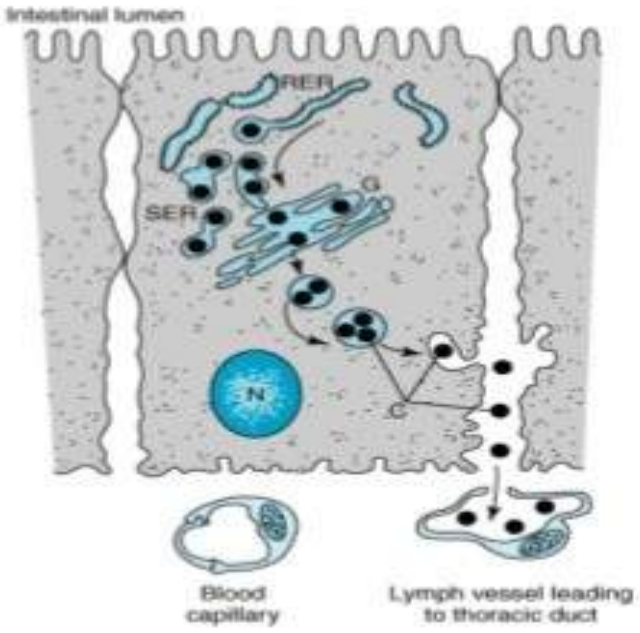
Some apolipoproteins are integral and cannot be removed, whereas others can be freely transferred to other lipoproteins.

Classification of Lipoproteins



Lipoproteins with high lipid content will have low density, larger size and so float on centrifugation. Those with high protein content sediment easily, have compact size and have a high density.

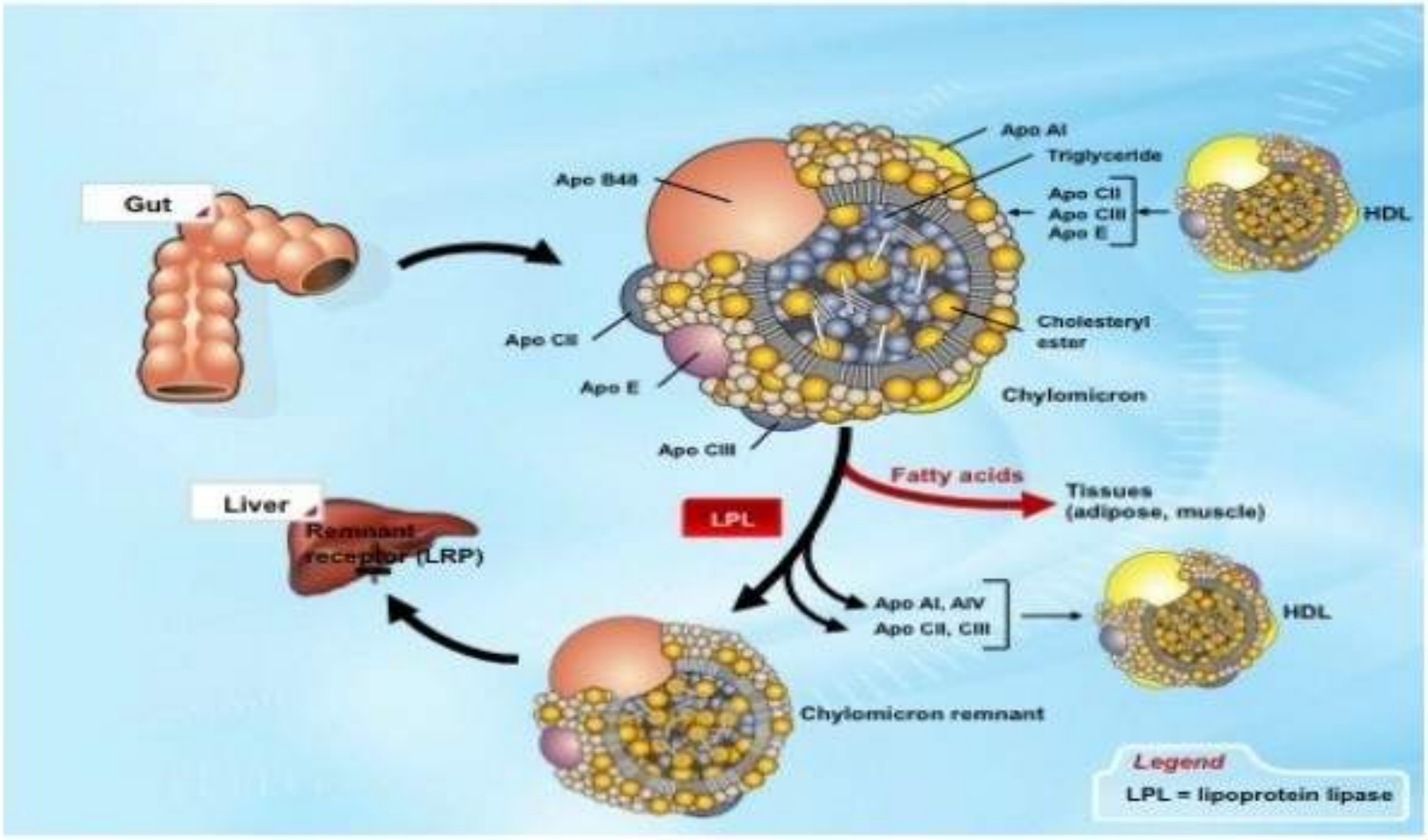
Metabolism of chylomicrons



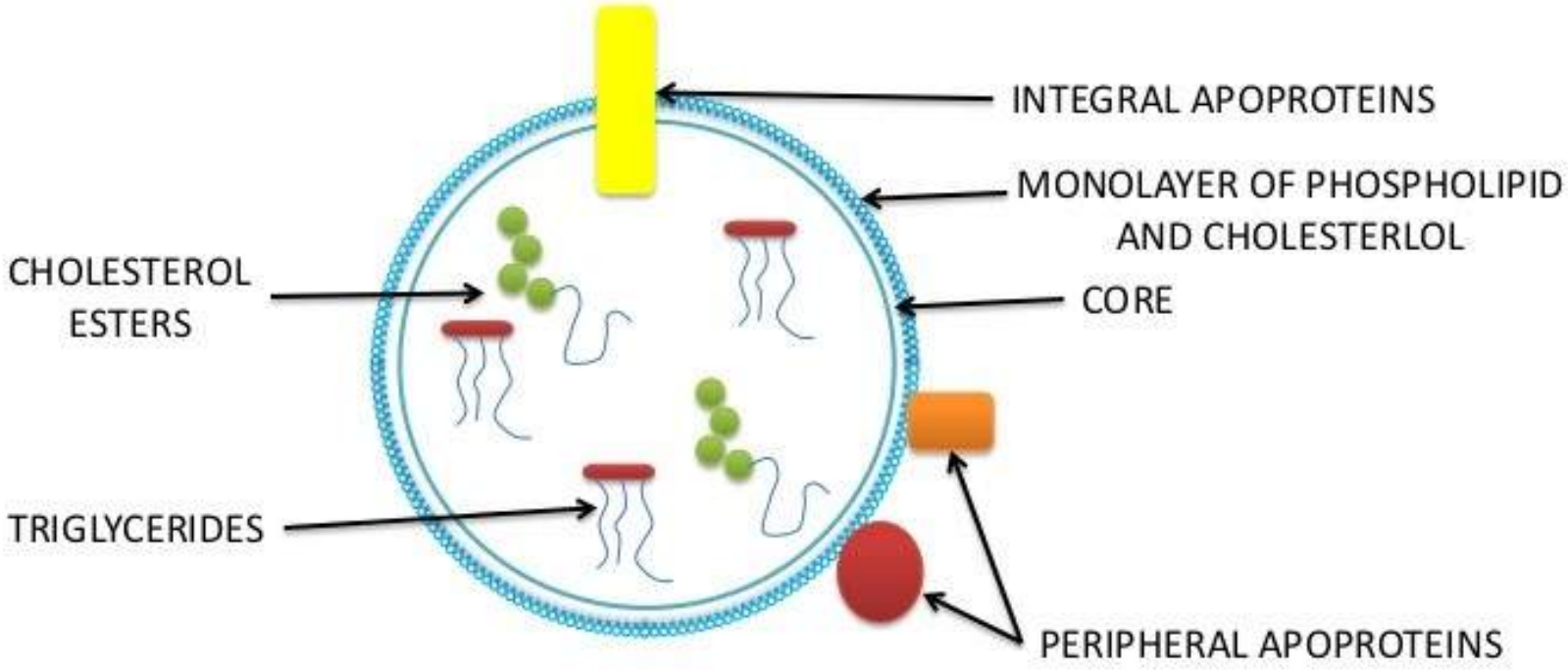
- Apolipoprotein B, synthesized in the RER, is incorporated into lipoproteins in the SER, the main site of synthesis of triacylglycerol.
- After addition of carbohydrate residues in G, they are released from the cell by reverse pinocytosis.
- Chylomicrons pass into the lymphatic system.

Clinical Significance- In **Abetalipoproteinemia** (a rare disease), lipoproteins containing apo B are not formed and lipid droplets accumulate in the intestine and liver(Due to non formation of VLDL)

Catabolism of Chylomicrons



Structure of lipoprotein



Nomenclature of lipoproteins

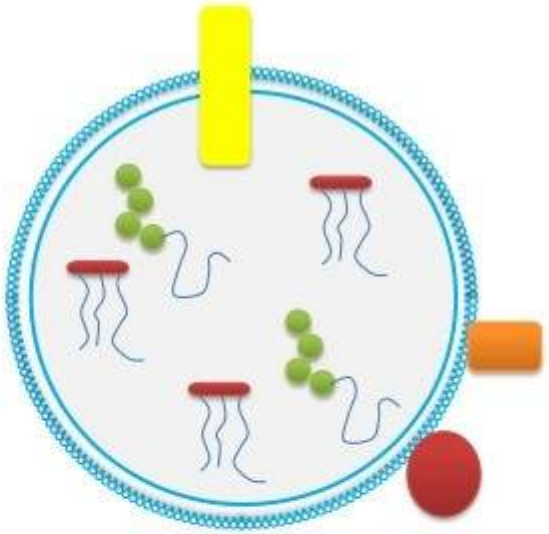


	Diameter nm	Protein %	Triglycerides %	Cholesteryl esters %
Chylomicron	75-1200	1	88	3
VLDL	30-80	10	56	15
IDL	25-30	10	29	34
LDL	18-25	~20	13	48
HDL	5-12	~50	13	30

Albumin Fatty acid complex

Apoproteins:

- Regulate the metabolism of lipoproteins by modulating the activity of enzymes
- Mediate the receptor mediated uptake of lipoproteins



Two kinds

1. Integral proteins
   - Apo-A
   - Apo-B (
2. Peripher
   - Apo- C
   - Apo- E

Name	Lipoprotein	Function
apoC-I	HDL, VLDL, IDL	Activator of LCAT
Name	**Lipoprotein**	**Function**
apoE	HDL, VLDL, IDL Chylomicrons	Binds to receptor on the cell membrane of liver
apoD	HDL, Chylomicrons nascent	Cholestreyl ester exchange

Beta oxidation

Oxidation of fatty acids is a major energy source in many organisms

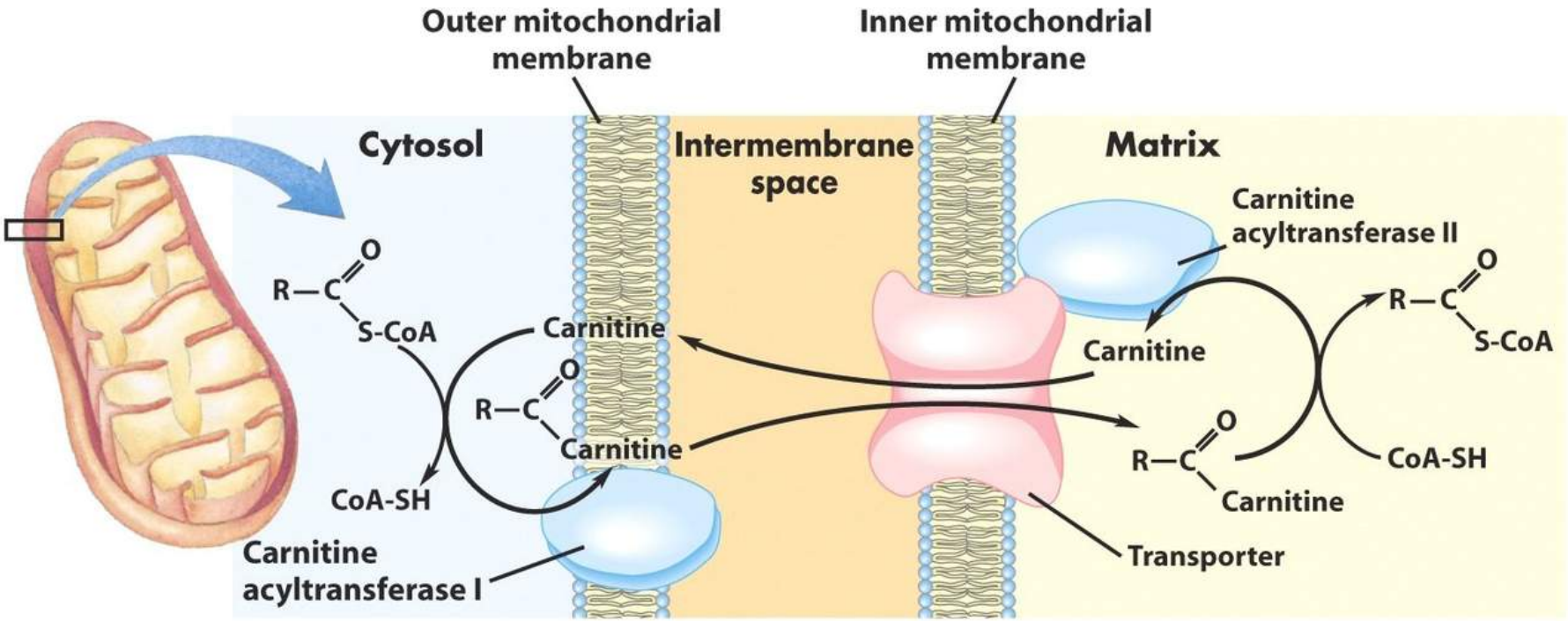
- About one-third of our energy needs comes from dietary triacylglycerols
- About 80% of energy needs of mammalian heart and liver are met by oxidation of fatty acids
- Beta oxidn is the main pw for FA
- It consists of repeated cycles of series of reactions
- 2 C atoms are removed from FA in each cycle
- Thus complete oxidation of palmitic acid (16 C) yields 8 mol

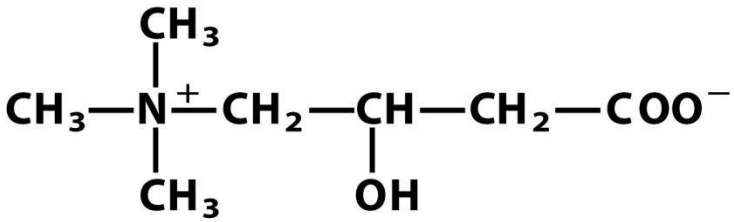
3 stages of beta oxidn

- 1 activation of FA
- FA + ATP → fattyacyl AMP + PPi
- Fattyacyl AMP + HSCoA → FattyacylCoA + PPi

- Transport of fattyacylCoA into mitochondria
- Fatty acyl CoA is produced in the cytoplasm, whereas B-Oxdn occurs in the mito
- Fattyacyl CoA is impermeable to i.m.m.
- Entry into mito needs a carrier called carnitine

Major Fatty Acid Oxidation = β-Oxidation Occurs in the Mitochondrion

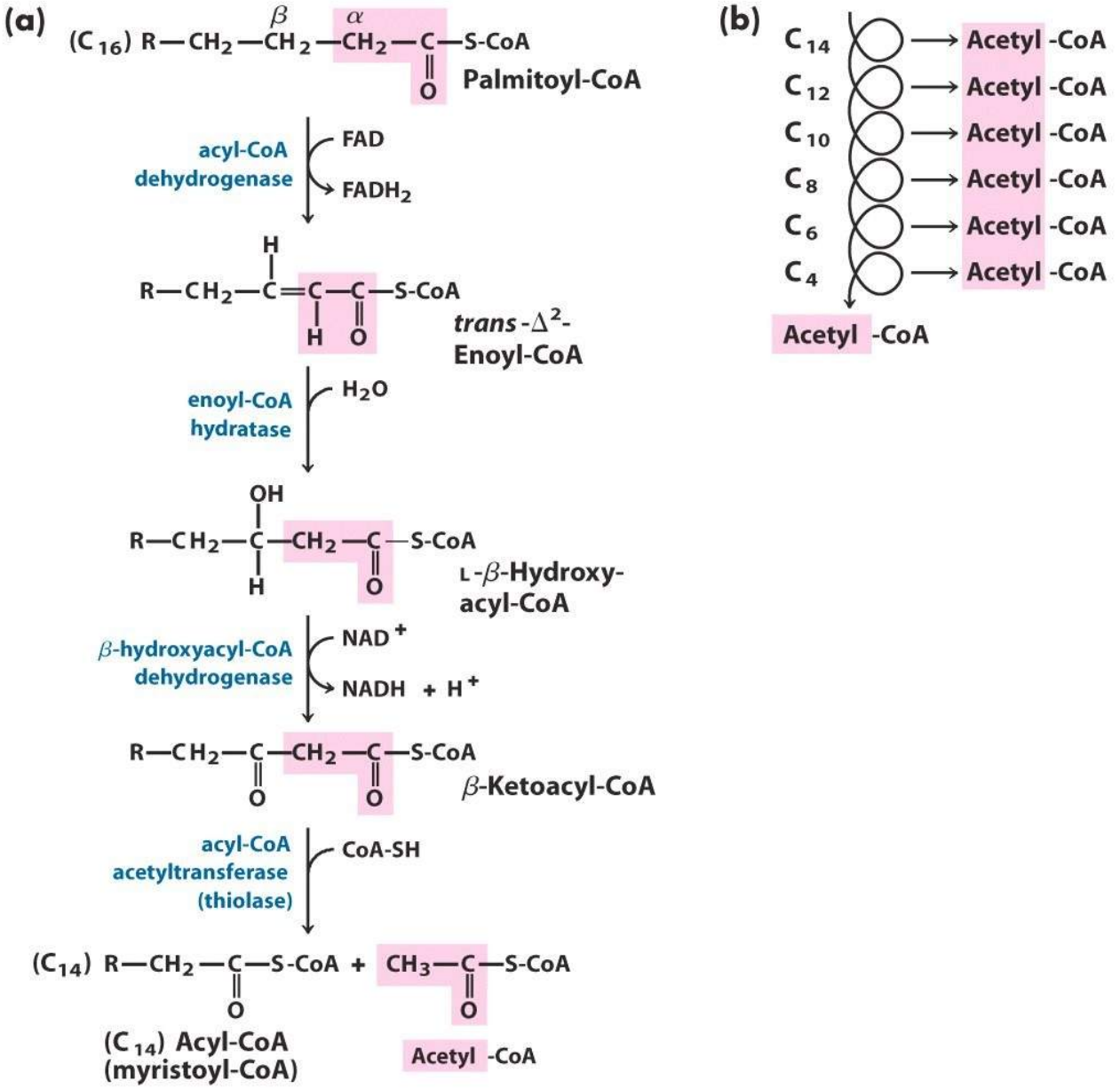




Carnitine

Unnumbered 17 p651
Lehninger Principles of Biochemistry, Fifth Edition
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One Round (a) and Further Rounds (b) of β-Oxidation



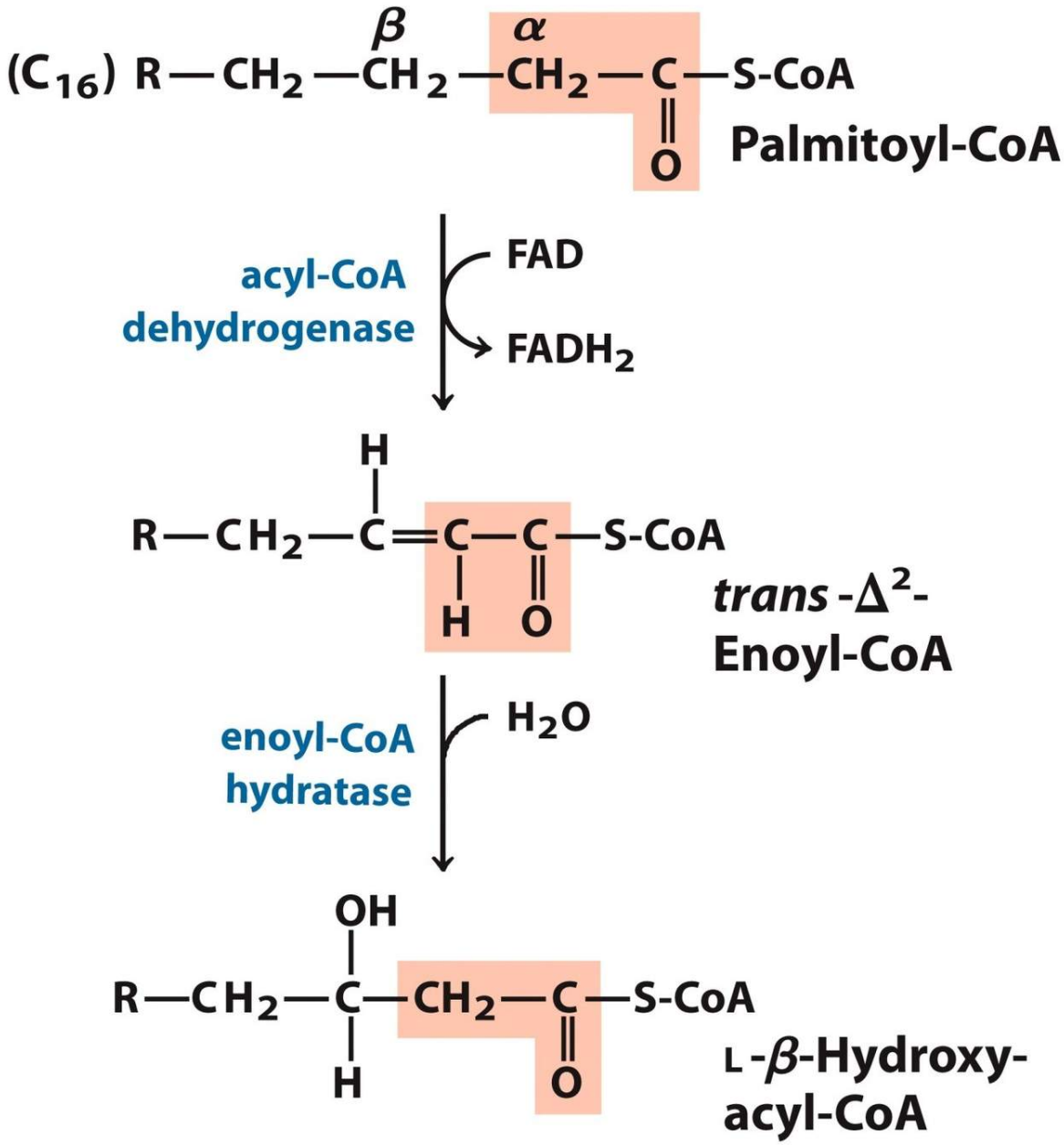


Figure 17-8a part 1
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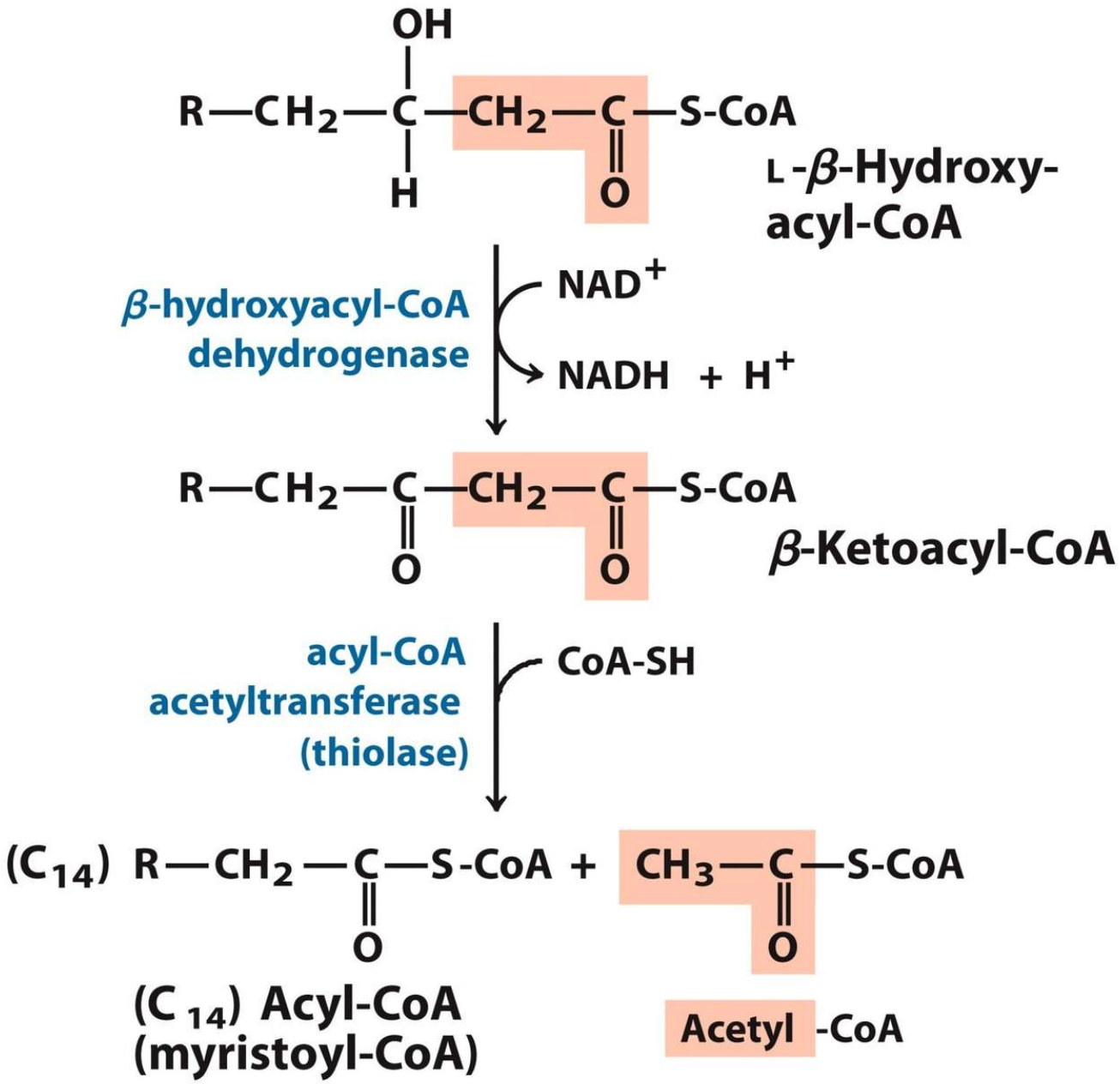


Figure 17-8a part 2
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- AcylCoA carnitine transferase == outer side of imm
- Translocase == integral membrane protein
- acylCoA carnitine transferase II == innerside of imm

Summary of B-oxdn of palmitic acid

- Palmitic acid + ATP + HSCoA → palmitoylCoA + AMP + PPi

- PalmitoylCoA + 7 FAD + 7 NAD + 7 H2O + 7 HSCoA →

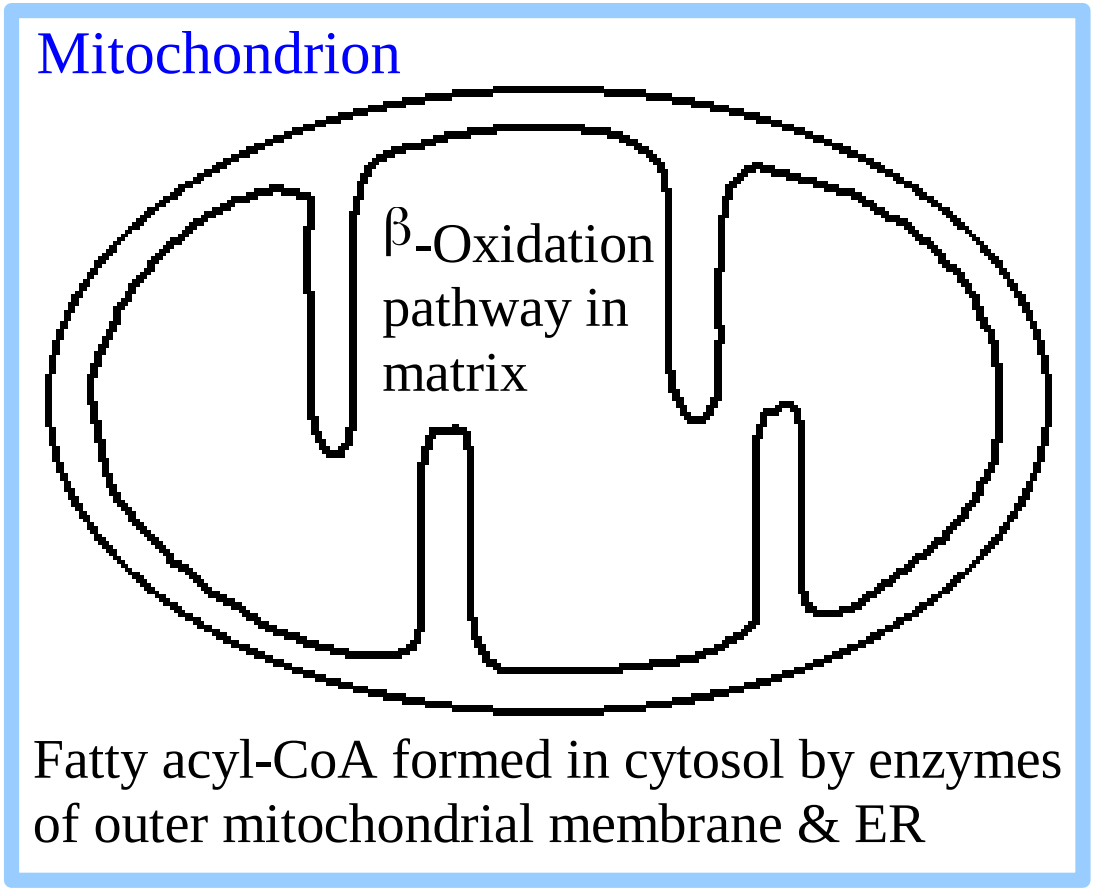
- 8 acetyl CoA + 7 FADH + 7 NADH + 7 H+

- Lippincott– Biochemistry- Illustrated Review

- THANK YOU

β-Oxidation pathway:

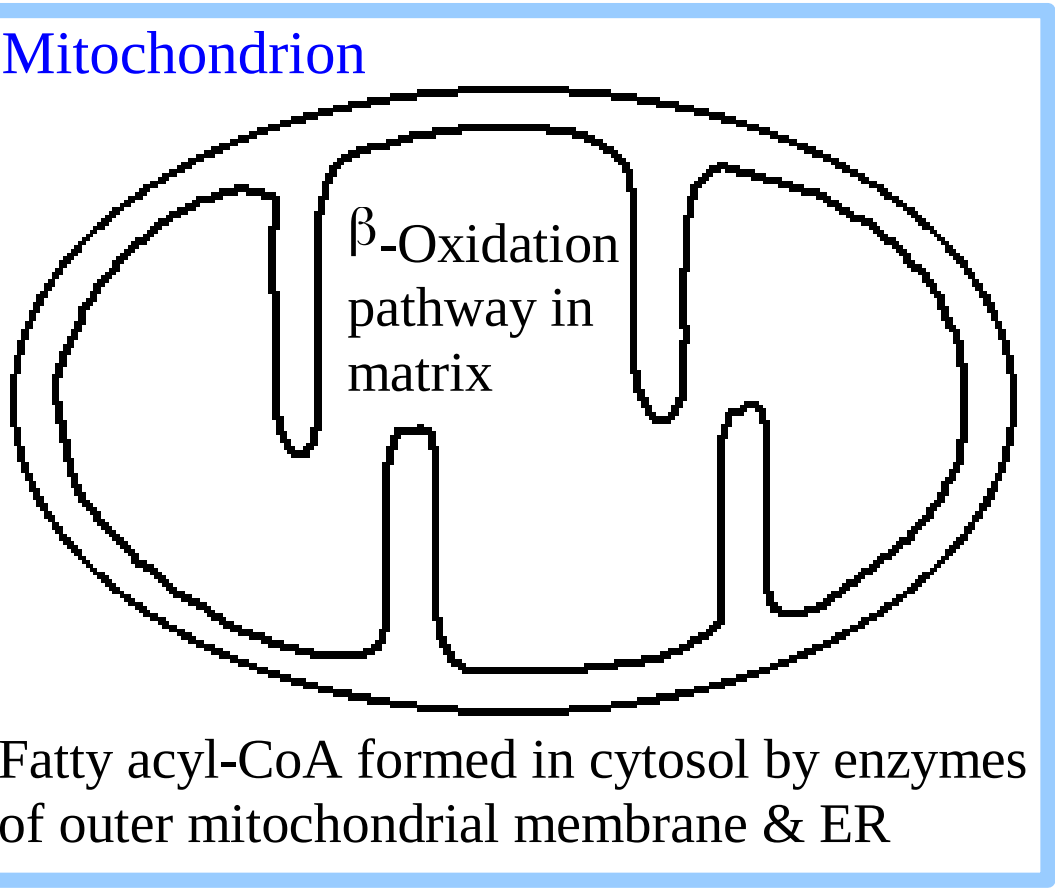
Fatty acids are degraded in the **mitochondrial matrix** via the β-Oxidation Pathway.



For most steps of the pathway there are multiple enzymes specific for particular fatty acid chain lengths.

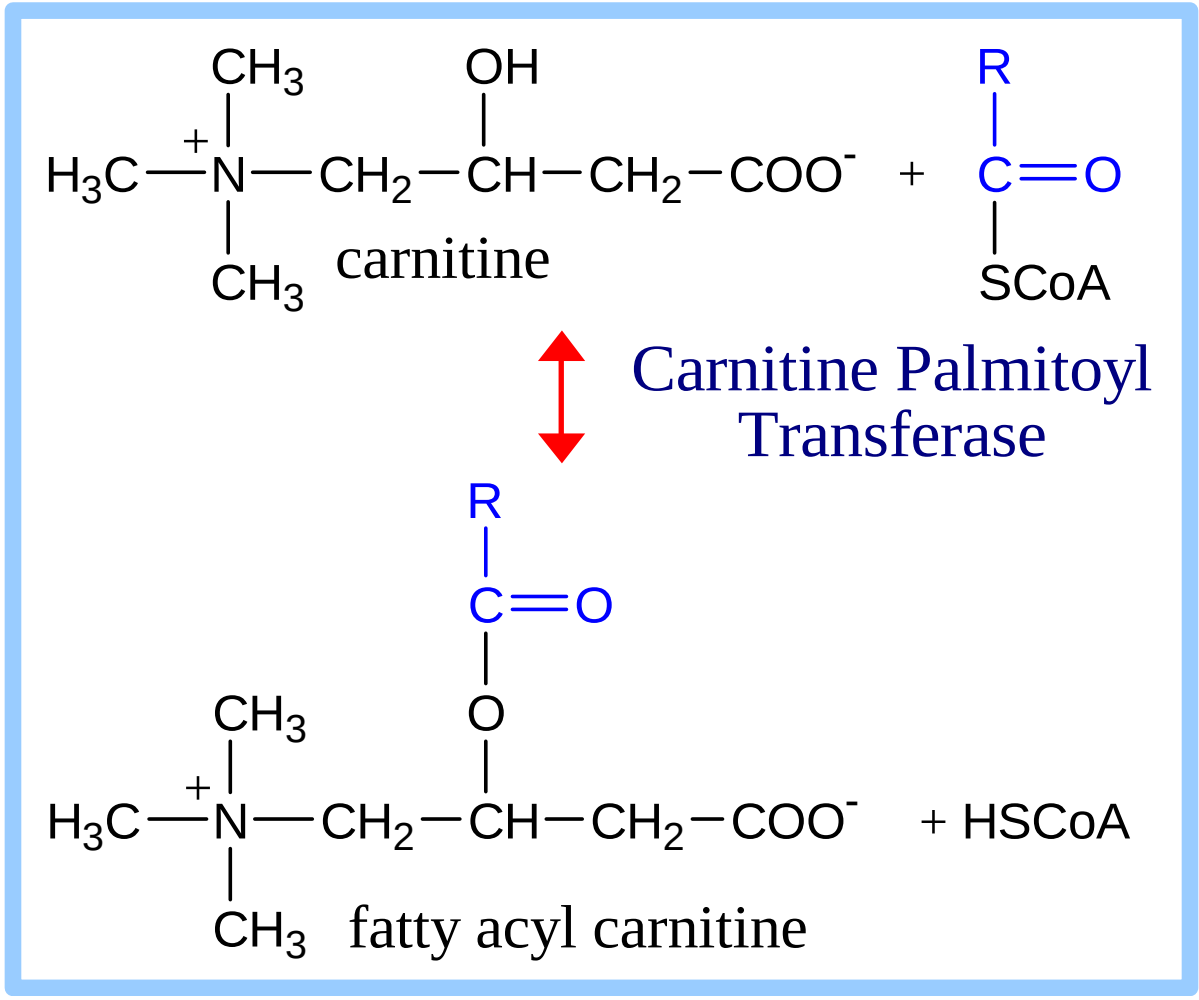
Many of the constituent enzymes are soluble proteins located in the mitochondrial **matrix**.

But enzymes specific for **very long chain fatty acids** are associated with the **inner membrane**, facing the matrix.



Fatty acyl-CoA formed outside can pass through the outer mitochondrial membrane (which has large VDAC channels), but cannot penetrate the inner membrane.

Transfer of the fatty acid moiety across the mitochondrial inner membrane involves **carnitine**.



Carnitine Palmitoyl Transferases catalyzes transfer of a fatty acid between the thiol of Coenzyme A and the hydroxyl on carnitine.